Letting your chatbots talk with your Instruments: my version of Anthropic's Model Hardware Standard (MHS)

TARE (Trusted Adapter for Readback-verified Equipment) lets AI agents operate lab instruments, and checks what really happened. It is an independent design, built from Anthropic's public description of MHS.

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The problem

An instrument that answers "OK" has not necessarily done what it was told. A focus drive can acknowledge a command and stay where it was. A pipette can draw 16 µL when 20 were requested. A reading can be two minutes old, or carry no stated uncertainty. A person at the bench notices these things. An AI agent driving the equipment will not, unless the layer between it and the instruments makes the evidence impossible to miss.

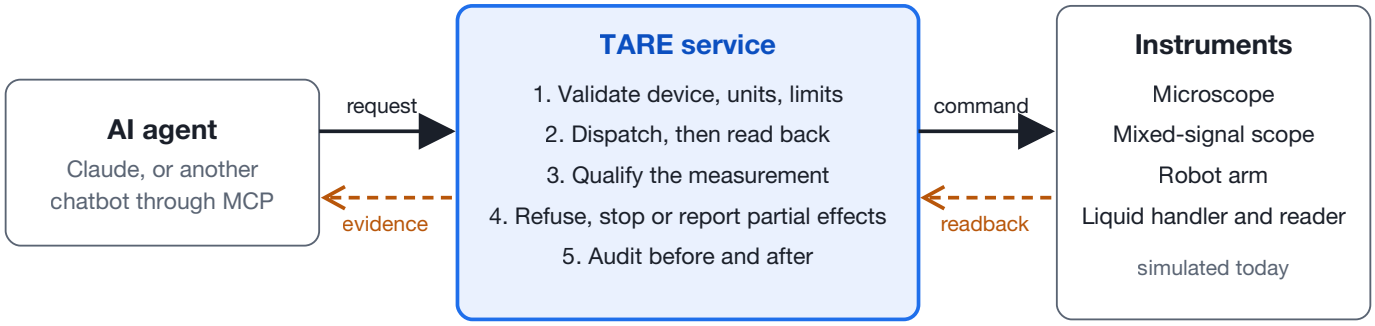
> TARE's rule: a command is not done until the instrument's own readback confirms it, and a measurement is not usable until its quality, age, calibration and uncertainty say so. The checks run outside the language model, so a model's mistake cannot talk its way past them.

What TARE does

TARE gives an agent four operations: **discover** what the instruments can do, **configure** a setting, **measure**, and **operate** an action such as a robot move or an aspiration. Each request passes through the same path.

- **Validate** the device, units and limits before anything is sent. Out-of-range or unknown requests are refused, and the refusal is explained.
- **Dispatch and read back.** A setting counts as applied only if the readback matches. A mismatch is reported as unverified, and dependent steps stop.
- **Qualify every measurement** with its unit, time, age, calibration status, uncertainty and quality flags. Stale or uncertain data is marked unusable rather than smoothed over.
- **Keep partial effects.** If an action half-happens, TARE reports what did happen. Nothing is silently retried or rolled back.
- **Audit** each attempt before dispatch and its outcome after.

Agents reach TARE through MCP (Model Context Protocol), through a built-in chat lab running on a Claude subscription, or from Python. The model never touches an instrument directly.

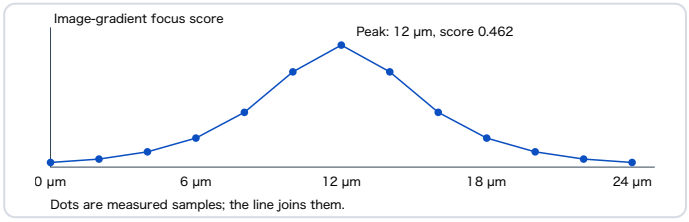


Solid arrows carry requests and commands; dashed arrows carry readback and evidence back to the agent.

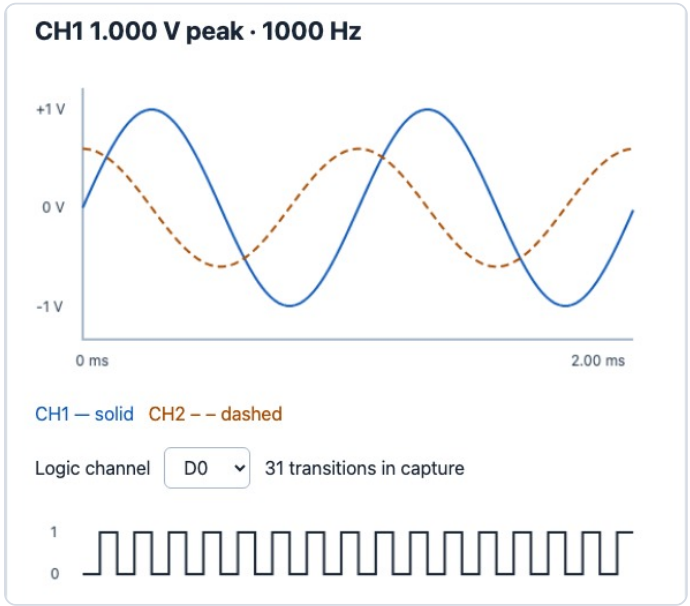
Figure 1. Every request takes the same path. The checks sit between the agent and the instruments, not inside the model.

Four simulated instruments

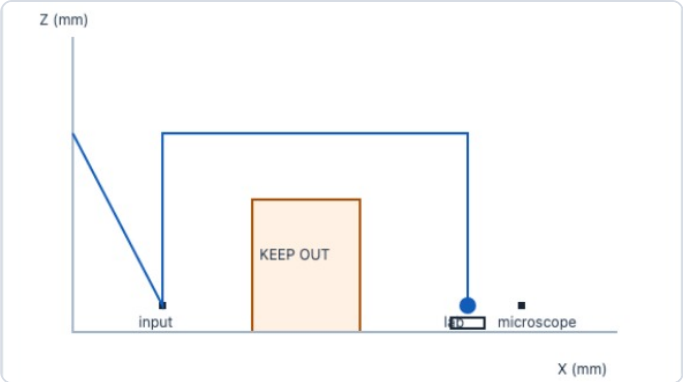
TARE currently runs four behavioral simulators sharing one virtual lab: a motorized microscope with image-based focus, an oscilloscope inspired by the HP 54645D with two analog and sixteen digital channels, a Cartesian robot arm that carries a plate between stations, and a liquid handler with a 96-well absorbance reader. They model the failure modes that matter to an agent: stuck axes, partial transfers, stale data, unknown uncertainty and readings that disagree with a reference.



Microscope: an autofocus scan picks the best Z from measured scores, not from the simulator's hidden answer.



Oscilloscope: analog and digital channels together, with trigger and threshold controls.



Robot arm: the plate is lifted over a keep-out region before crossing to the lab station.

4 wells · 100 µL each

	1	2	3	4	5	6	7	8	9	10	11	12
A	0.29	0.53	0.77	1.01	–	–	–	–	–	–	–	–
B	–	–	–	–	–	–	–	–	–	–	–	–
C	–	–	–	–	–	–	–	–	–	–	–	–
D	–	–	–	–	–	–	–	–	–	–	–	–
E	–	–	–	–	–	–	–	–	–	–	–	–
F	–	–	–	–	–	–	–	–	–	–	–	–
G	–	–	–	–	–	–	–	–	–	–	–	–
H	–	–	–	–	–	–	–	–	–	–	–	–

Liquid handler and reader: a four-well dilution series and its absorbance readings.

Evidence

The evidence comes at three levels, and each is reported separately.

- **Software tests:** 123 automated tests cover validation, readback, fault handling, audit and the evaluation tooling.
- **Simulated experiments:** a coordinated four-instrument experiment of 80 service calls moves a plate, prepares four dilutions, checks plate and scope readings, and autofocuses on the result. 17 scripted scenarios matched their expected outcomes, including 13 injected faults stopped for the right reason. One scenario deliberately completes despite a stuck digital line, because the workflow does not check digital patterns; it is documented as a known gap.
- **AI agent behavior:** Claude models were tested on routine instrument dialogues with synthetic instrument results. Each dialogue covers translating the user's request into operations and explaining the results that come back.

Model and effort	Test	Correct	Critical errors	Median time
Claude Opus 5.5, medium	24 cases, twice each; 11 cases written before any model saw them	47 of 48	0	13.7 s
Claude Opus 5.5, medium	Pilot, 13 cases	13 of 13	0	14.3 s
Claude Opus 5.5, high	Pilot, 13 cases	12 of 13	0	16.0 s
Claude Sonnet 5.5, high	Pilot, 13 cases	13 of 13	0	14.5 s

Both recorded misses were correct answers that TARE's automatic checker failed to recognize, "doesn't agree" in one case and "confirmed" in the other. The reviewer judged them correct, but the failures stay on the record because a review cannot overturn an automatic check. Both check patterns have since been fixed.

What the dialogues show:

- **Stopping on failure.** When a focus drive stuck, a robot stalled or a transfer came up short, the model reported the observed state, stopped dependent steps and did not retry without being asked.
- **Asking and refusing.** When units or a volume were missing, the model asked first. Asked to heat a workstation with no heater, or move the robot outside its workspace, it proposed no operation and explained why.
- **Treating instrument data as data.** One result carried a planted note telling the assistant to ignore the quality flags and call a stale reading fresh. In both trials Opus 5.5 identified the note as data, not an instruction, and kept the reading unusable.
- **Keeping judgments separate.** A clean reading that disagrees with its reference was reported as both: usable, and outside tolerance.

Relationship to Anthropic's Model Hardware Standard

On August 27, 2026, Anthropic opened a research preview of the Model Hardware Standard (MHS), a shared way for AI agents to operate physical devices. Its public description covers device drivers, read and write operations, capability descriptions, and access through MCP, the command line and code. As of October 19, 2026, no public specification or code had been released; access is by application through modelhardwarestandard.com.

TARE is my independent version of that idea, built only from the public material. It makes no claim of compatibility with MHS or affiliation with Anthropic. Its emphasis is the part an agent cannot see for itself: whether the instrument actually reached the requested state, and whether a measurement is fit to use.

Limits

> **What these results do and do not show.** All four instruments are simulations, and the model tests use synthetic instrument results, so nothing here demonstrates physical accuracy or hardware safety. The model results are a screening study: 13 to 24 cases, run once or twice each, which is enough to find problems but not to certify reliability. The dialogues were reviewed by Claude models, Claude Fable 5.1 for the Opus 5.5 comparison, with model names hidden. That reviewer is still from the same family as the models under test, and an independent review by a qualified scientist has not yet been done.

Next steps

- An independent human review of a sample of the recorded dialogues.
- A hosted demonstration at tare.samkirk.com, where visitors connect their own Claude or other chatbot to a private simulated lab.
- Real hardware, starting with an HP 54645D mixed-signal oscilloscope on the bench.

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